SERVICE MANUAL

Invivo Corporation
3545 SW 47th Avenue
Gainesville, FL 32608

Document TR0228S

Revision 12

GE 1.5T 8 Channel Neurovascular Array

GE Catalog Part Number: M3087JB

Invivo Part Number: 800228

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

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- Call 1–800–548–3366 and use option 8.
- Fill out a report on http://egems.med.ge.com/edq/home.jsp
- Contact your local service coordinator for more information on this process.

Rev. June 13, 2006

LANGUAGE POLICY FOR SERVICE DOCUMENTATION (DOC0620154 REV 1)

ПРЕДУПРЕЖДЕНИЕ (BG)

Това упътване за работа е налично само на английски език.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告 (ZH-CN)

本维修手册仅提供英文版本。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
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警告 (ZH-HK)

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- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
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- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE (HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

VÝSTRAHA (CS)

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL (DA)

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING (NL)

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING
(EN)

This service manual is available in English only.

- If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

HOIATUS
(ET)

See teenindusjuhend on saadaval ainult inglise keeles

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS
(FI)

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION
(FR)

Ce manuel d’installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d’installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG
(DE)

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ
(EL)

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS
(HU)

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése.
- Ne próbálja elkezdeni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN
(IS)

Þessi þjónustuhandbók er aðeins fáanleg á ensku.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA
(IT)

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告
(JA)

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고
(KO)

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS
(LV)

Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS
(LT)

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL
(NO)

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE
(PL)

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

AVISO
(PT-BR)

Este manual de assistência técnica encontra-se disponível unicamente em inglês.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

ATENÇÃO
(PT-PT)

Este manual de assistência técnica só se encontra disponível em inglês.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

ATENȚIE
(RO)

Acest manual de service este disponibil doar în limba engleză.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării şi înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!
(RU)

Данное руководство по техническому обслуживанию представлено только на английском языке.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

UPOZORENJE
(SR)

Ovo servisno uputstvo je dostupno samo na engleskom jeziku.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja servisera, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

UPOZORNENIE
(SK)

Tento návod na obsluhu je k dispozícii len v angličtine.

- Ak zákazníkov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

ATENCION
(ES)

Este manual de servicio sólo existe en inglés.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING
(SV)

Den här servicehandboken finns bara tillgänglig på engelska. .

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DİKKAT
(TR)

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

TABLE OF CONTENTS

SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

This is a service manual for the GE SIGNA 1.5T 8 Channel Neurovascular Array.

NOTE: Coils do not ship with phantoms. Phantoms come in a unifed phantom set with the MR system.

SHIPPING LIST – TABLE 1-1

Description	GE Part #	INVIVO #	Qty
1.5T 8 Channel Neurovascular Array	2317115-2	103045	1
Operator Manual	2317115-4	500078	1
Service Manual	2317115-5	TR0228S	1
NV Array Positioner	5344806	N/a	1
Wedge Pad	2317115-8	101407	2
HNC Pad	2317115-6	102955	1

1-2 Compatibility

This coil is compatible with the following hardware configurations

- Signa Horizon and LX Horizon 1.5T with Excite Technology (8-channel)
- Excite II (11.x), Excite III (12.x), HDx/HDxt/HDi that are upgrades.

1-3 Related Documentation

- Signa 11.x Service Methods CD, 2333500-1
- 8 Channel Neurovascular Array Operator's Manual, 2317115-4

1-4 Environmental Requirements

Storage Requirements

Coil should be stored flat in the scanner room.

Dimensions

Coil Dimension:	724.0 mm x 457.0 mm x 427.0 mm	(28.50 in x 18.00 in x 16.80 in)

Weight

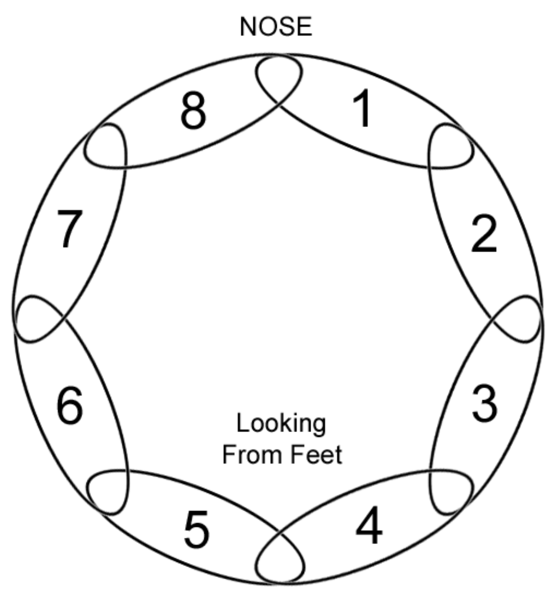
Coil Weight:	9.675 Kg	(21.5 lb)

1-5 Theory of Operation

Refer to Figure 1 and the Block Diagram.

The 1.5T 8 Channel Neurovascular Array consists of 13 elements that are divided into 2 sections, the head section and the neck section. The head section consists of 8 elements. Each element is comprised of a loop arranged azimuthally around the head. Each loop is approximately 45° in azimuthal extent. The center of Loop #1 is located 22.5° rotated in the clockwise direction (as viewed from the feet) with respect to the nose. Each of the next seven loops is rotated clockwise by 45° successively. Loops #1 and #8 are located anteriorly while loops #4 and #5 are located posteriorly. Loops #2 and #3 are located on the patient Left side while Loops #6 and #7 are located on the patient Right side. During transmit, each loop is decoupled with both passive and active decoupling, with an additional active preamp protection network. During receive, signals from 1 – 8 are directed to separate ultra-low-input impedance preamp modules placed directly at the output of the loop, providing additional element decoupling.

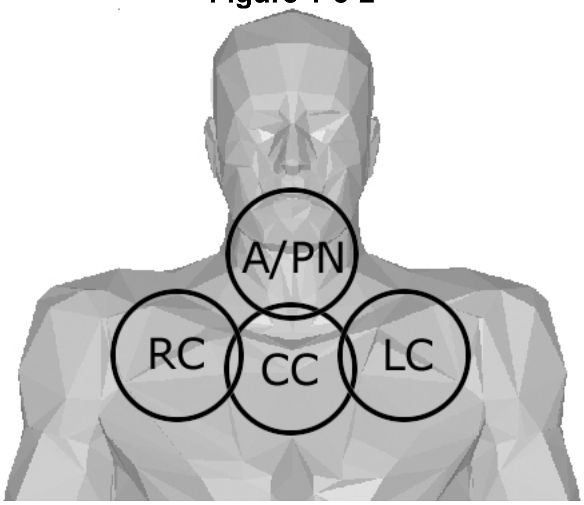
Figure 1-5-1



Loop Element Diagram

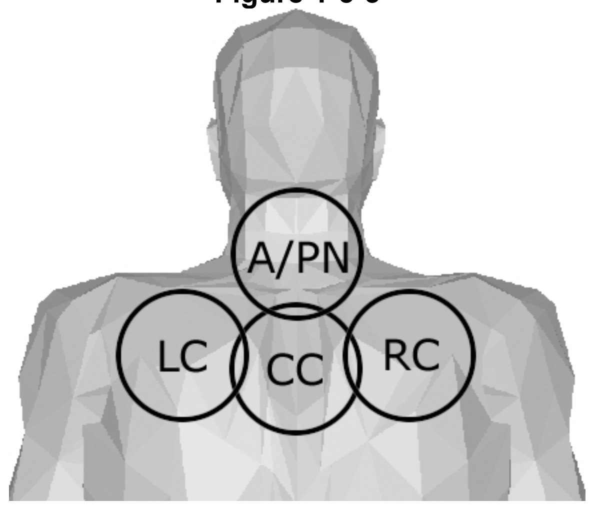
The neck section consist of 5 elements, arranged as detailed in Figures 2 through 5. Table 1 describes the chest and neck element names.

Figure 1-5-2



Front Elements

Figure 1-5-3

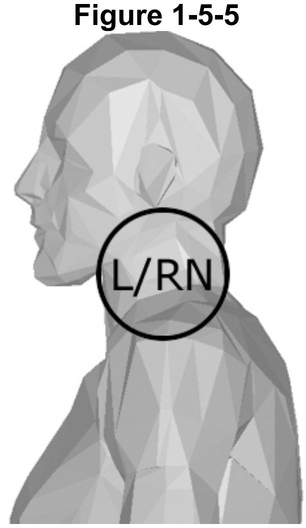


Back Elements

Figure 1-5-4

Right Side Element

Figure 1-5-5

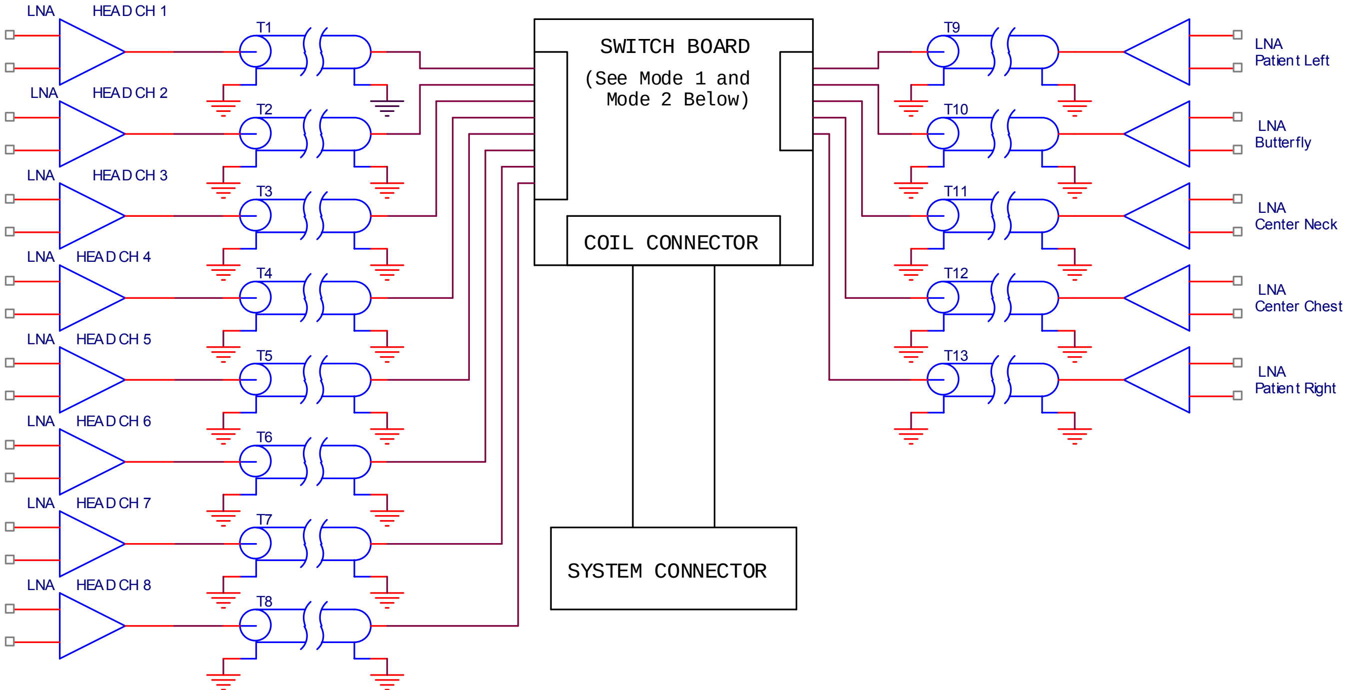


Left Side Element

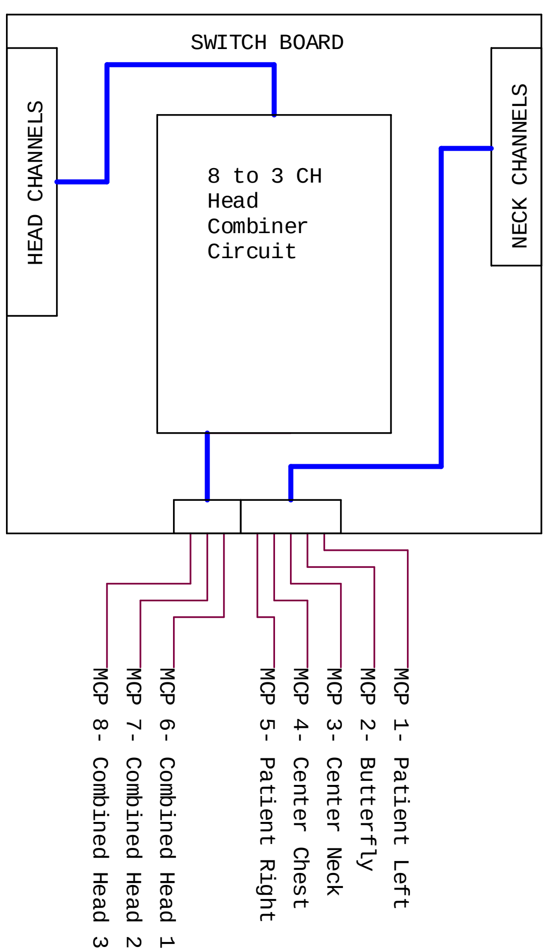
RC	Right Chest
LC	Left Chest
A/PN	Anterior/Posterior Neck
CC	Center Chest
L/RN	Left/Right Neck

There are two modes of operation that utilize the two coil sections. Mode one is the head only mode where each of the 8 head elements is directed to the output. Mode two utilizes a switchable 8-to-3 combiner at the output of the head section and the five elements in the neck section for a total of 8 signals at the output. See the block diagram on the following page for more detail.

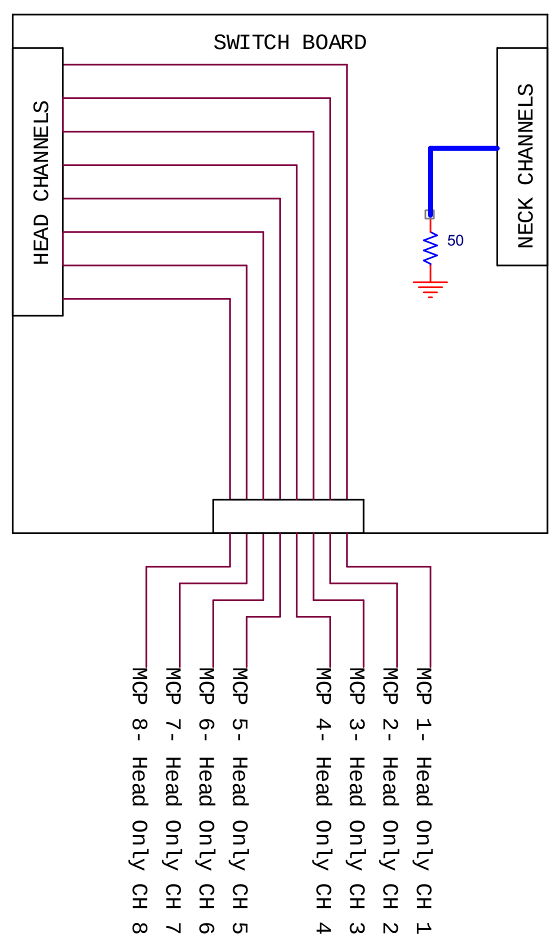
1.5T 8 Channel Neurovascular Array Block Diagram



MODE 1



MODE 2



SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

1. The mirror assembly is installed by sliding [in the superior direction] the grooved section of the mirror assembly into the grooved section atop the anterior coil. To remove, press the release button on the mirror assembly while sliding the mirror assembly in the inferior direction.

2. The upper mirror is designed to "break away" should it come into contact with the bore. The upper mirror can be reset by rotating it back into position until the latches are secured.

3. Remove the protective sheet of plastic from the surface of each mirror.

4. The 8 Channel NV Coil requires 10.0 M4 or greater software.

5. In order to use the 8 Channel NV coil, sites with 10.0 M4 or greater software require the installation of the 8 Channel NV Array QA Tool (2376441 rev. SW002). See direction 2382643 for procedure, as found in the shipping contents of the coil and the Service Methods CD.

2-1-2 Installing the Coil

2-1-2-1 Configuration File Installation

In order to use the 8 Channel NV coil, config files on the system must be updated.

1. Using the config file manager, remove all currently installed configs for the 8 Channel NV array by MRI Devices (Invivo).

2. Using the config file manager, install the following coil names:

8NVHEAD_A	8NVARRAY_A
8NVHEAD ONE RCVR	8NVANGIO_A
8NVHEAD NECK_A	8NVANGIO LAT
8NVHEAD NECK	8NVANGIO MED
8NVHService	8NVAService

2-1-2-2 Customer Protocol Modification

Update the system protocol files using CoilCorrect.sh. See section 9-2 of "Host Software Install Release 10.X", direction OW2SCA8G, as found on the Service Methods CD. Translate the config names as shown in Table 2-1-2-2.

CONFIG NAMES TRANSLATION CHART – TABLE 2-1-2-2

10.0 M4 Name	New Name
8NVHEAD	8NVHEAD_A
8NVHEAD ONE RCVR	8NVHEAD ONE RCVR
8NVHEAD LONG	8NVHEAD NECK_A
8NVHEAD LONG 3	8NVHEAD NECK
8NVARRAY	8NVARRAY_A
8NVARRAY 7	8NVANGIO_A
8NVANGIO	8NVANGIO LAT
8NVANGIO 3	8NVANGIO MED

2-1-2-3 System Reboot and Save Info

1. Reboot the system to allow all coil configuration changes to take place.

2. Perform a “save info” to preserve the new settings in case they need to be reloaded in the future.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

2. Perform Section 3 – Coil Imaging Performance Verification.

2-3 Customer Notification

1. Leave the customer information kit, 2385392-100 and customer letter 2393321 at the site or deliver to the customer.

2. Send “8NVArraySupport@med.ge.com” confirmation that the coil has been installed. Provide the system ID, customer contact name, and phone number in the e-mail.

2-4 Periodic Quality Assurance Check

On a periodic basis, such as during installation and planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Check plastic screws for tightness, damage, or missing.
3. Perform Section 3-2 – Coil Imaging Performance Verification and record data values in Data Sheet (Table 7-1).
4. If SNR is trending downwards, perform a system level signal to noise check. Also check the interconnect pins for damage or abuse.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED USING LEGACY PHANTOM – TABLE 3-2-1

Description	GE Part #	Qty
Legacy 8 Ch NV Phantom Kit	2317115-3	1

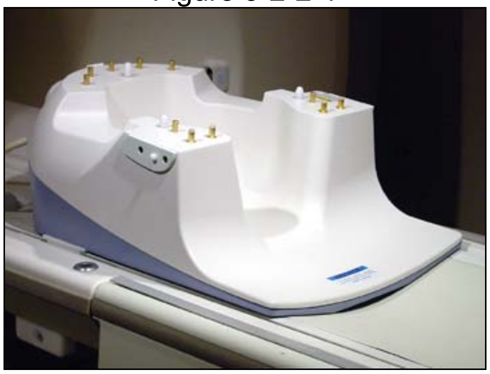
TOOLS REQUIRED USING UNIFIED PHANTOM – TABLE 3-2-2

Description	GE Part #	Qty
17cm TLT Head Spherical Phantom	46-265826G6	1
10cm Spherical Phantom	46-317586G1	1
Large Cylindrical unified phantom	5342679	1
Phantom positioner, NV	5344806	1

3-2-2 QA Phantom and Coil Setup with Legacy Phantom

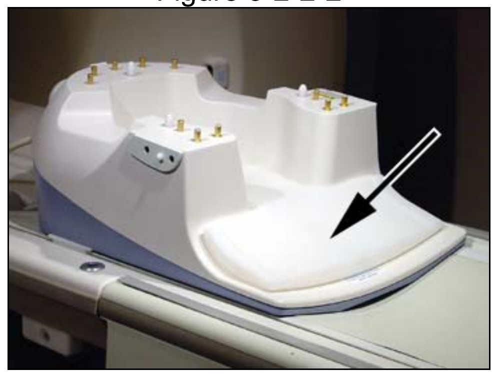
1. End the previous exam if this has not already been done.
2. Attach the base of the 8 channel neurovascular array to the cradle in the same position on the cradle as the standard head coil as shown in Figure 3-2-2-1 and remove the top.
3. Note the serial number of the coil.
4. Place phantom positioner on coil as shown in Figure 3-2-2-2.
5. Place phantom on positioner as shown in Figure 3-2-2-3.
6. Attach top to coil as shown in Figure 3-2-2-4.
7. Establish an axial landmark on the line on top of the coil. Advance to scan.

Figure 3-2-2-1



Remove Coil Top

Figure 3-2-2-2



Place Phantom Positioner on coil

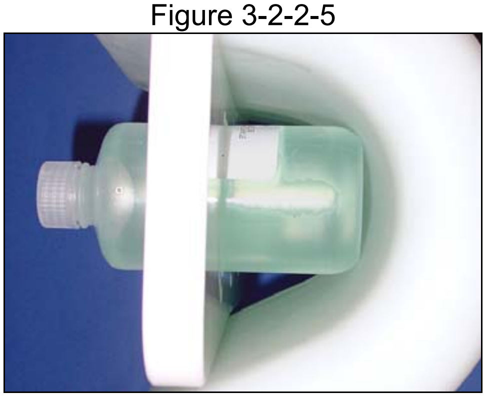
Figure 3-2-2-3

Place Phantom on Positioner

Figure 3-2-2-4

Place Top on Coil

Figure 3-2-2-5



Correct Bottle Placement

NOTE: For correct bottle placement, insert the bottom of each phantom bottle fully, until it touches the curved inside surface of the main phantom. Refer to Figure 3-2-2-5.

3-2-3 QA Phantom and Coil Setup with Unified Phantom

Visual	Description
 **Figure 3-2-3-1**	1. Place the base of the NV array coil on the patient table and latch it to the table as shown in Figure 3-2-3-1.
 **Figure 3-2-3-2**	2. Place the phantom positioner in the coil as shown in the Figure 3-2-3-2.
 **Figure 3-2-3-3**	3. Place the phantoms in the positioner as shown in the Figure 3-2-3-3.
	4. Carefully align the connector pins on the base of the Coil with the top part of the coil and latch both the coil sections into place as in Figure 3-2-3-4. (The scanner will not operate if the coil sections are not latched

Figure 3-2-3-4	correctly).

3-2-4 MCQA / SNR Test

Continue for 14x Software Versions or Greater	Continue for 11x and 12x Software Versions
1. On the Common Service Desktop, select Image Quality, and then the MCQA tool. 2. Make the proper coil selection and press the “Start Scan” button. 3. Further information on how to run the MCQA tool can be found on 14x/15x service CD or on web at: Troubleshooting/Image Quality/multicoil Quality Assurance.	1. From the service browser, select Image Quality **->** Coil SNR **->** <u>Click here to start this tool</u> 2. Select the coil type you are testing, enter the serial number and select Start Scan. **Note:** Additional documentation can be accessed by selecting File **->** Help **->** 8NV **Note:** The SNR tool requires that **8NVAservice** and **8NVHservice** service configuration files be installed. The image quality check uses two consecutive acquisitions for the same slice. The signal is measured at the center of the phantom with a ROI of about 60% of the phantom size. The noise is measured on the subtracted image of the two consecutive signal images using the same ROI. SNR is calculated by dividing the Signal mean by the noise standard deviation. All measurements are performed on composite images. Interim images are acquired for reference only. **Proceed to TABLE 3-2-6-2 SNR LIMITS**

3-2-5 SNR Analysis 11.x and 12.x only

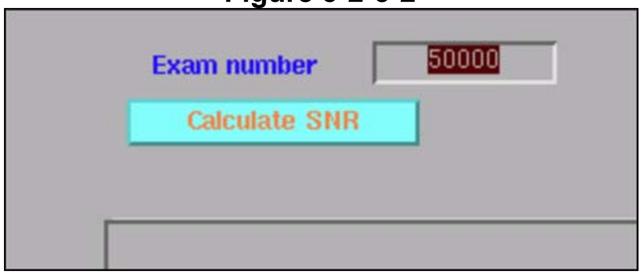
1. In the command window enter **snr_for_8nvarray_QA** and press enter. A screen like that shown in Figure 3-2-5-1 should appear.

Figure 3-2-5-1 SNR Analysis Screen

2. In the Exam number window, enter the exam number as shown in Figures 3-2-5-2 and 3-2-5-3 (The Delete key must be used instead of Backspace).

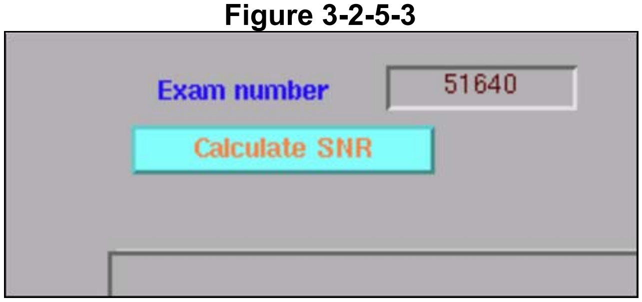
3. In the Coil serial number window, enter serial number and bay number as shown in Figures 3-2-5-4 and 3-2-5-5 (Optional).

Figure 3-2-5-2



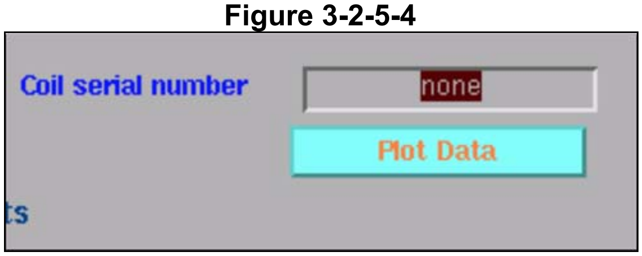
Highlight Digits for Exam Number

Figure 3-2-5-3



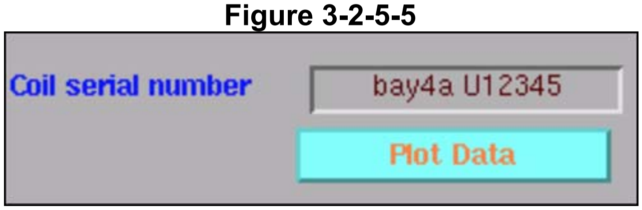
Type In Exam Number

Figure 3-2-5-4



Highlight Coil Serial Number

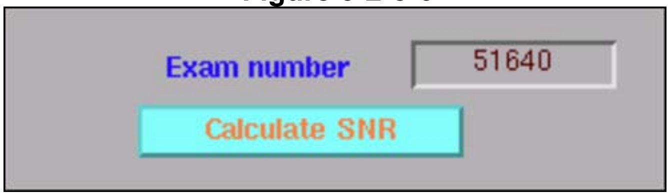
Figure 3-2-5-5



Enter Coil Serial Number and Bay Number

4. Click the Calculate SNR button as shown in Figure 3-2-5-6.

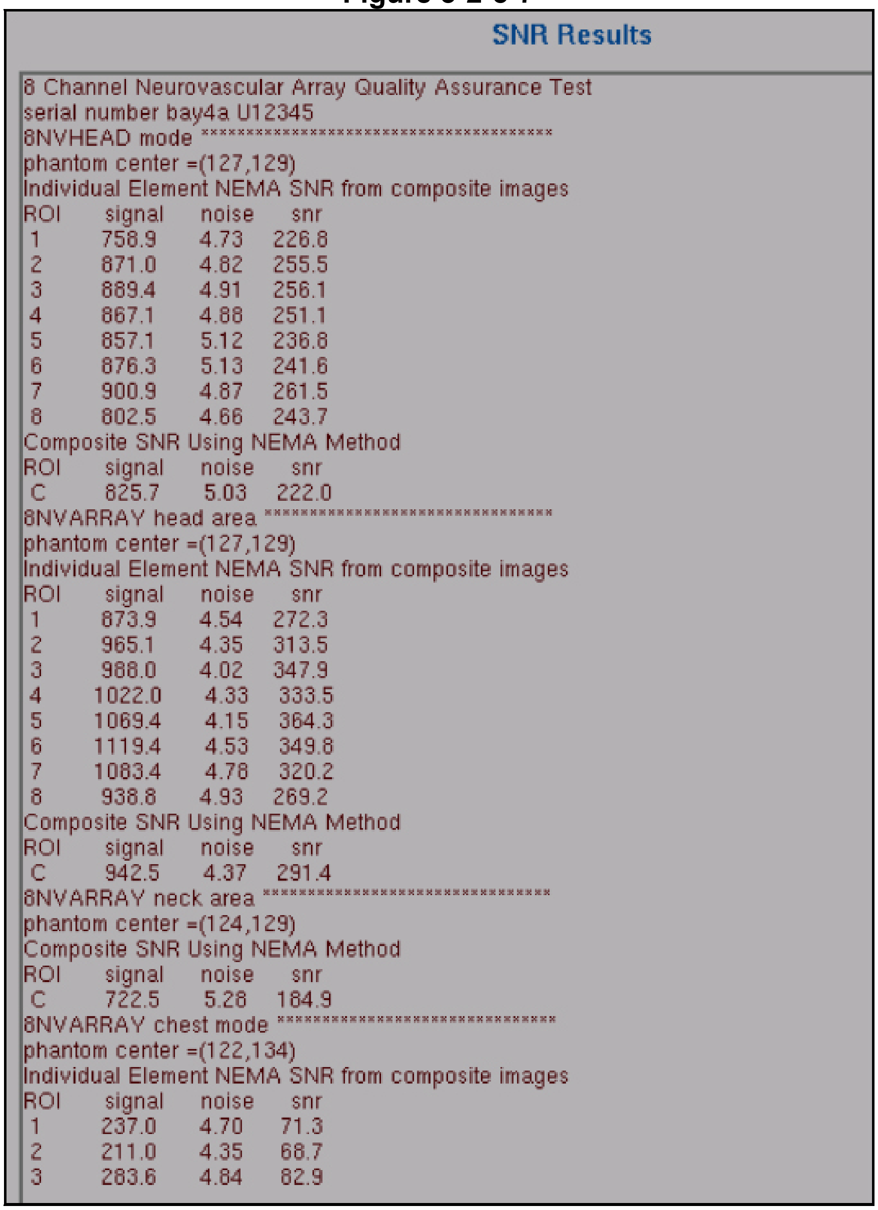
Figure 3-2-5-6



Click Calculate SNR Button

5. Results will appear as shown in Figure 3-2-5-7.

Figure 3-2-5-7



SNR Results

8 Channel Neurovascular Array Quality Assurance Test
serial number bay4a U12345
8NVHEAD mode **
phantom center =(127,129)
Individual Element NEMA SNR from composite images

ROI	signal	noise	snr
1	758.9	4.73	226.8
2	871.0	4.82	255.5
3	889.4	4.91	256.1
4	867.1	4.88	251.1
5	857.1	5.12	236.8
6	876.3	5.13	241.6
7	900.9	4.87	261.5
8	802.5	4.66	243.7

Composite SNR Using NEMA Method

ROI	signal	noise	snr
C	825.7	5.03	222.0

8NVARRAY head area ********************************
phantom center =(127,129)
Individual Element NEMA SNR from composite images

ROI	signal	noise	snr
1	873.9	4.54	272.3
2	965.1	4.35	313.5
3	988.0	4.02	347.9
4	1022.0	4.33	333.5
5	1069.4	4.15	364.3
6	1119.4	4.53	349.8
7	1083.4	4.78	320.2
8	938.8	4.93	269.2

Composite SNR Using NEMA Method

ROI	signal	noise	snr
C	942.5	4.37	291.4

8NVARRAY neck area ********************************
phantom center =(124,129)
Composite SNR Using NEMA Method

ROI	signal	noise	snr
C	722.5	5.28	184.9

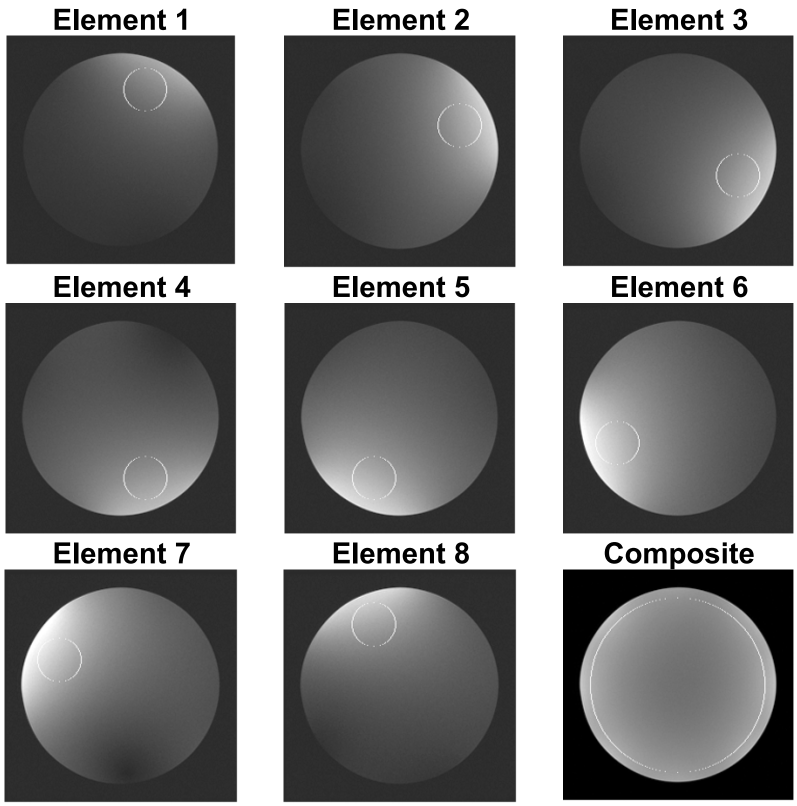
8NVARRAY chest mode ******************************
phantom center =(122,134)
Individual Element NEMA SNR from composite images

ROI	signal	noise	snr
1	237.0	4.70	71.3
2	211.0	4.35	68.7
3	283.6	4.84	82.9

SNR Results

6. Table 3-2-6-1 shows the test results that correspond to each test.

Figure 3-2-5-8



Individual Head Element ROIs

Note: The results are also stored in a file in the /usr/g/service/data directory. The filename will be similar to 8nvarray_51897_c7ga0522.coilqa where 51897 is the exam number.

7. Record the coil serial number and the associated filename in Table 3-2-5-3 for future reference.

   Note: The analysis tool can be used to process multiple runs by entering a new exam number and coil serial number and pressing the Calculate SNR button again. It is not necessary to exit the analysis tool after each run.

8. Proceed to the next section for instructions to plot the data.

3-2-6 Plot Results 11x and 12x only

Note: Perform steps 1 through 3 only if the SNR tool is not already running.

Note: Table 3-2-6-1 explains the output of the tool. Table 3-2-6-2 provides SNR acceptance limits. Table 3-2-6-3 provides a data sheet for recording file names for future reference.

1. Open a command window on the tools desktop.

2. Navigate to the /usr/local/bin/CoilsTools directory.

3. In the command window enter **snr_for_8nvarray_QA** and press enter. A screen like that shown in Figure 3-2-6-1 should appear.

4. Click on the plot data button as shown in Figure 3-2-6-1.

5. Click on the “List 8NVARRAY data files” button as shown in Figure 3-2-6-2.

6. Click on the name of the file to be plotted as shown in Figure 3-2-6-3.

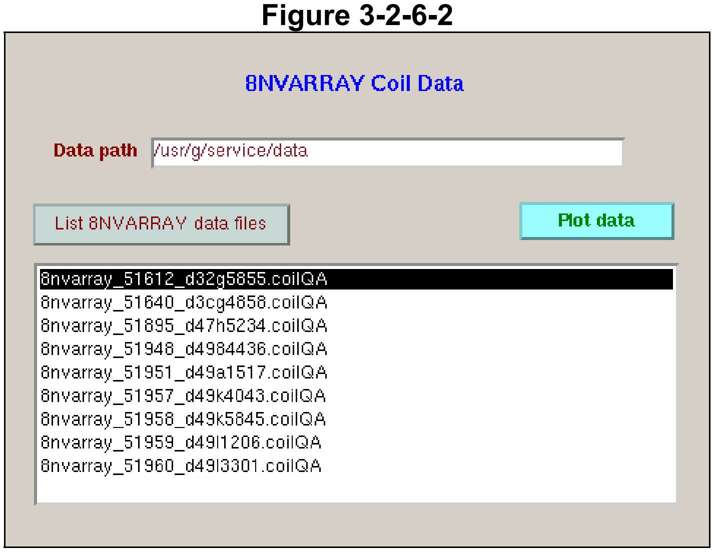
7. Click on the plot data button as shown in Figure 3-2-6-4 to process the file.

8. Example of a nominal coil is shown in Figure 3-2-6-5.

Figure 3-2-6-1

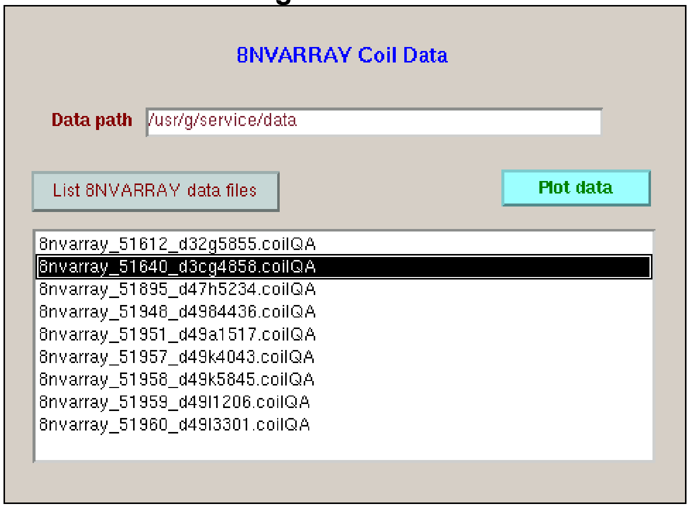
Start Plot Tool

Figure 3-2-6-2



List Files

Figure 3-2-6-3



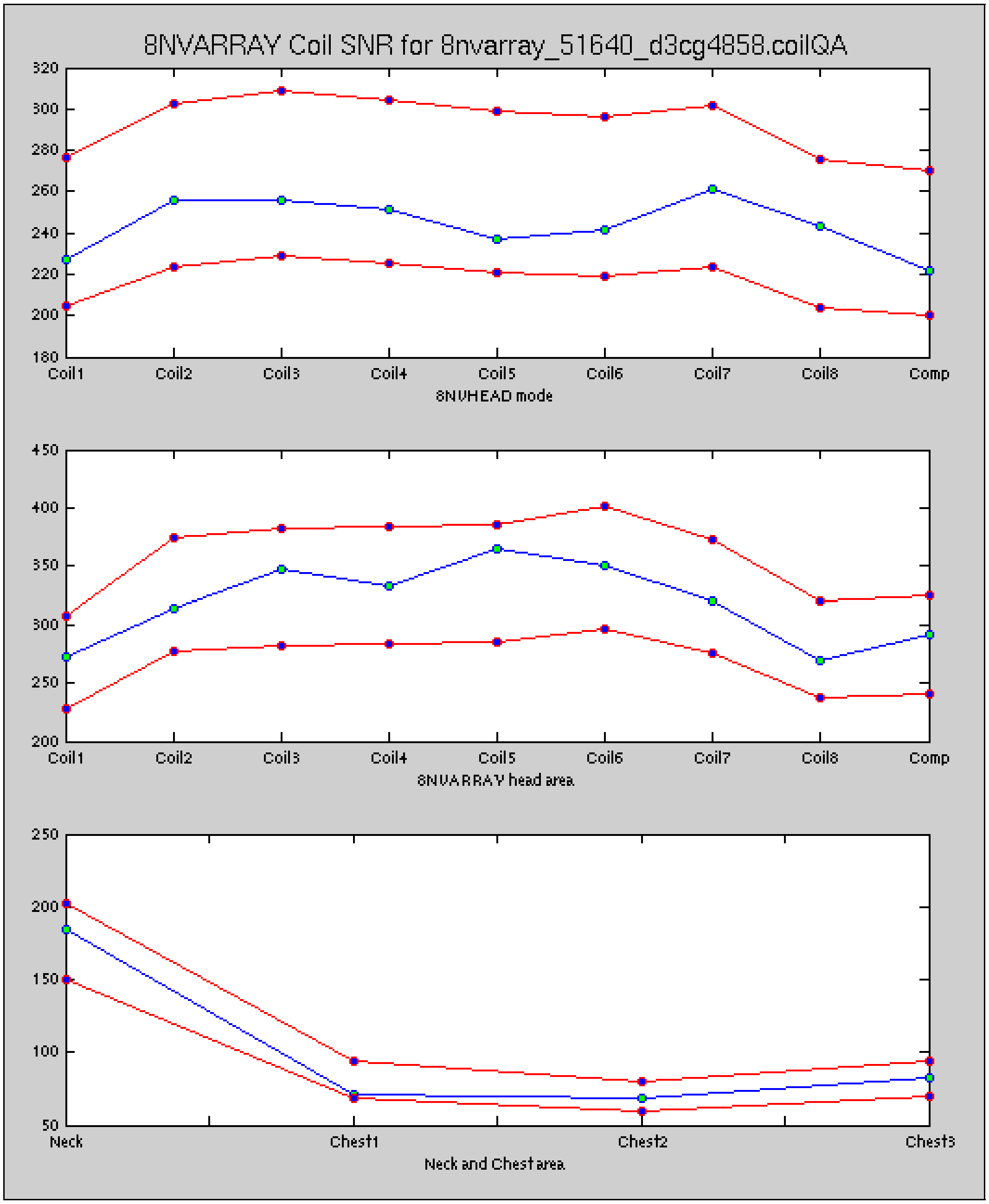
Select File

Figure 3-2-6-4

Process File

Figure 3-2-6-5

+



Results Compared to Upper and Lower Limits

Note: Red lines show upper and lower acceptance limits. Blue line shows test results.

TABLE 3-2-6-1 EXAMPLE OUTPUT OF TOOL

```
8 Channel Neurovascular Array Quality Assurance Test
serial number bay4a U12345
8NVHEAD mode ****************************************
phantom center =(127,129)
Individual Element NEMA SNR from composite images
ROI      signal     noise     snr
 1        758.9      4.73     226.8
 2        871.0      4.82     255.5
 3        889.4      4.91     256.1
 4        867.1      4.88     251.1
 5        857.1      5.12     236.8
 6        876.3      5.13     241.6
 7        900.9      4.87     261.5
 8        802.5      4.66     243.7
Composite SNR Using NEMA Method
ROI      signal     noise     snr
 C        825.7      5.03     222.0
8NVARRAY head area ********************************
phantom center =(127,129)
Individual Element NEMA SNR from composite images
ROI      signal     noise     snr
 1        873.9      4.54     272.3
 2        965.1      4.35     313.5
 3        988.0      4.02     347.9
 4       1022.0      4.33     333.5
 5       1069.4      4.15     364.3
 6       1119.4      4.53     349.8
 7       1083.4      4.78     320.2
 8        938.8      4.93     269.2
Composite SNR Using NEMA Method
ROI      signal     noise     snr
 C        942.5      4.37     291.4
8NVARRAY neck area ********************************
phantom center =(124,129)
Composite SNR Using NEMA Method
ROI      signal     noise     snr
 C        722.5      5.28     184.9
8NVARRAY chest mode *******************************
phantom center =(122,134)
Individual Element NEMA SNR from composite images
ROI      signal     noise     snr
 1        237.0      4.70      71.3
 2        211.0      4.35      68.7
 3        283.6      4.84      82.9
```

Using ROIs shown in Figure 3-2-2-8 signal is measured in each of the two composite images of the head area and averaged. The same ROI for each element is used to measure noise in the filtered difference image.

Composite image results

Using ROIs shown in Figure 3-2-2-8 signal is measured in each of the two composite images of the head area and averaged. The same ROI for each element is used to measure noise in the filtered difference image.

Composite image results

Composite image results

Composite image results measured on a circular ROI on each of 3 bottles in chest area.

TABLE 3-2-6-2 SNR LIMITS

			USL	Nominal	LSL
8NVHEAD mode	head	elem1	268	229	189
		elem2	301	257	213
		elem3	305	260	215
		elem4	309	264	218
		elem5	307	262	217
		elem6	313	267	221
		elem7	311	265	219
		elem8	275	235	194
		comp	270	231	191
8NVARRAY mode	head	elem1	327	279	231
		elem2	388	331	274
		elem3	407	348	288
		elem4	426	364	301
		elem5	429	366	303
		elem6	423	361	299
		elem7	394	336	278
		elem8	334	285	236
		comp	352	300	248
	neck		230	196	162
	chest 1		88	75	61
	chest 2		81	69	56
	chest 3		94	80	66

TABLE 3-2-6-3 DATA SHEET

Coil Serial Number	Filename

3-2-7 Example Images

Figure 3-2-7-1
Receiver 1
8NVHEAD Mode Localizer Image

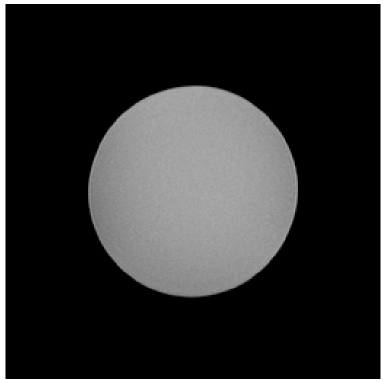


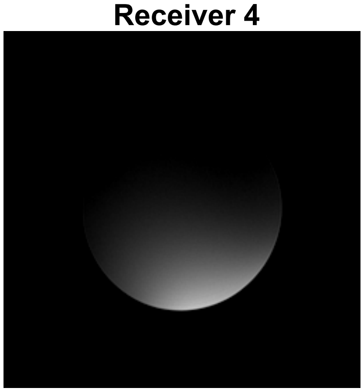
Figure 3-2-7-2 8NVHEAD Mode Signal Images

Receiver 1

Receiver 2

Receiver 3

Receiver 4



Receiver 5

Receiver 6

Receiver 7

Receiver 8

Composite

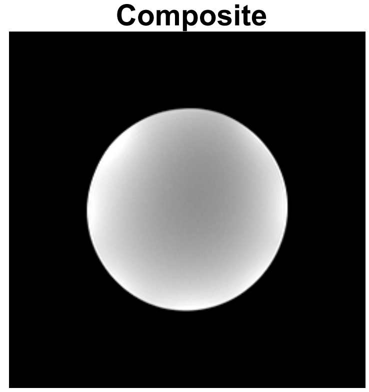


Figure 3-2-7-3 8NVARRAY Mode Localizer Images

Slice 1 (chest) Slice 2 (neck) Slice 3 (head)

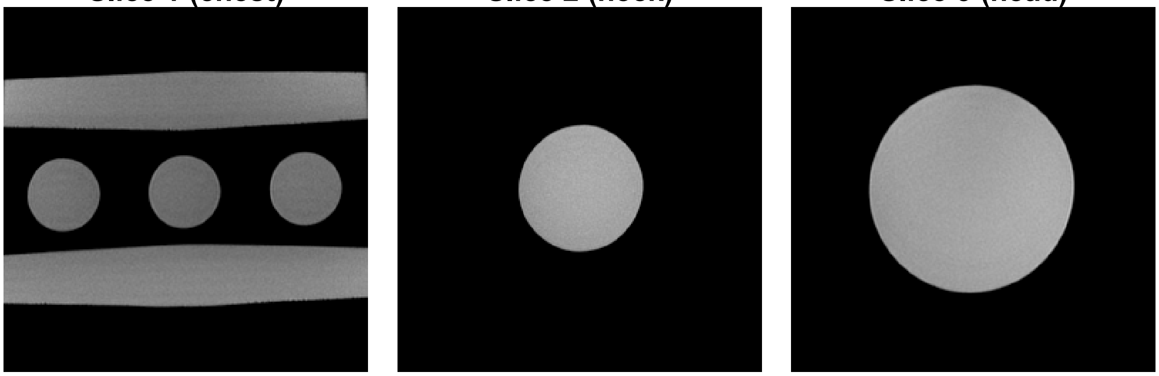
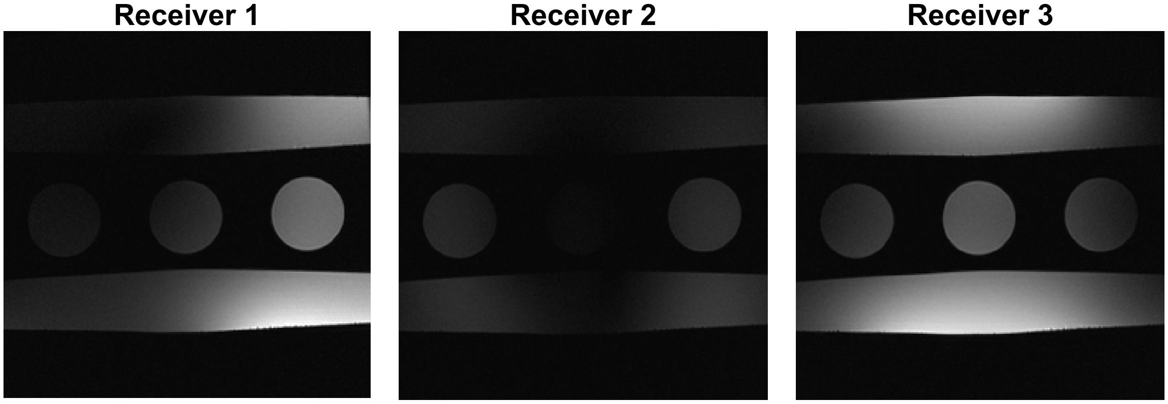
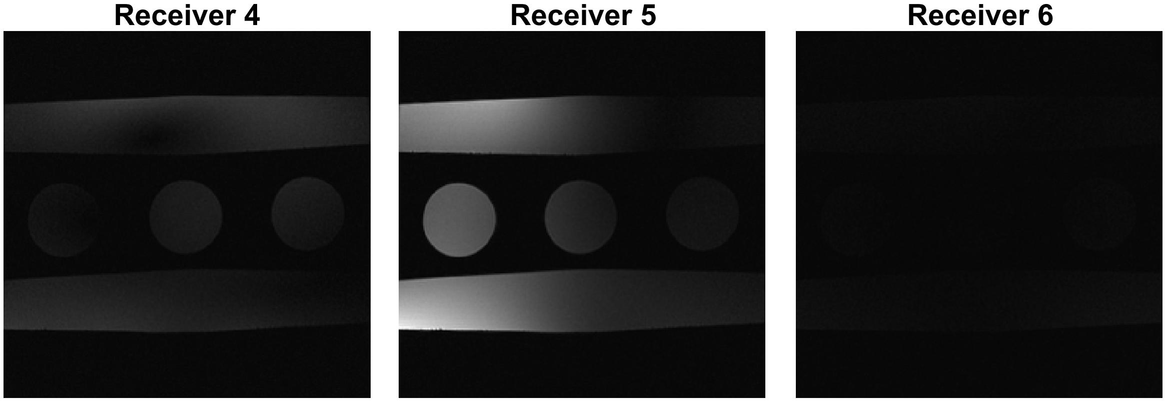


Figure 3-2-7-4 8NVARRAY Mode Signal Images Slice 1 (chest)

Receiver 1 Receiver 2 Receiver 3



Receiver 4 Receiver 5 Receiver 6



Receiver 7 Receiver 8 Composite

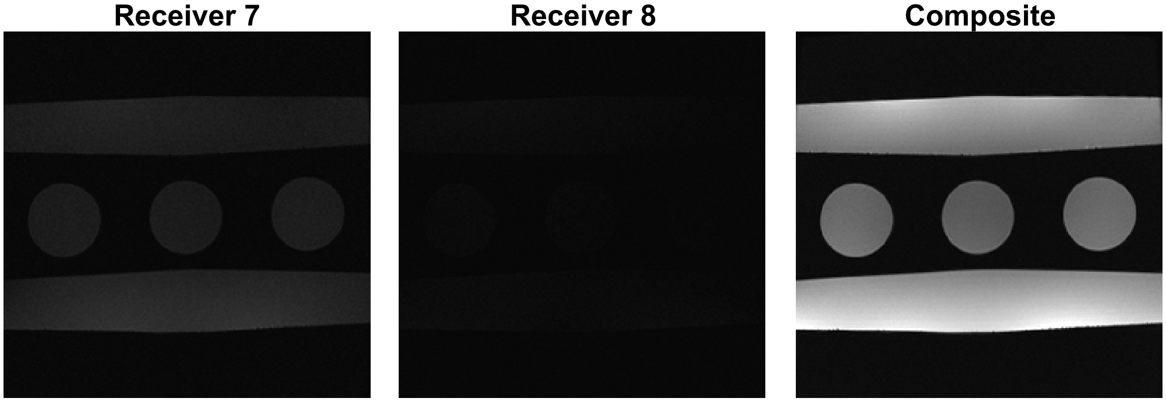


Figure 3-2-7-5 8NVARRAY Mode Signal Images Slice 2 (neck)

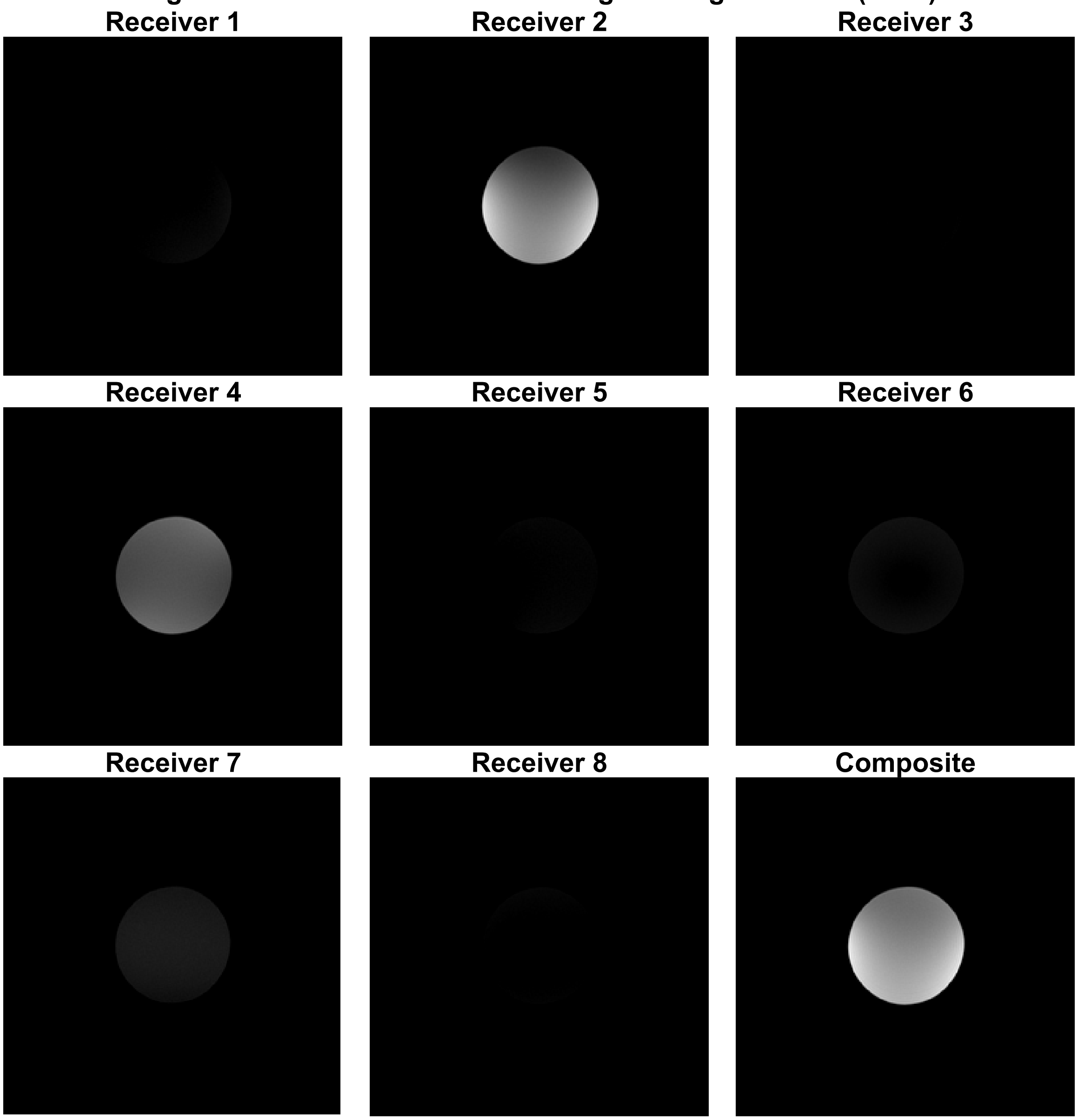
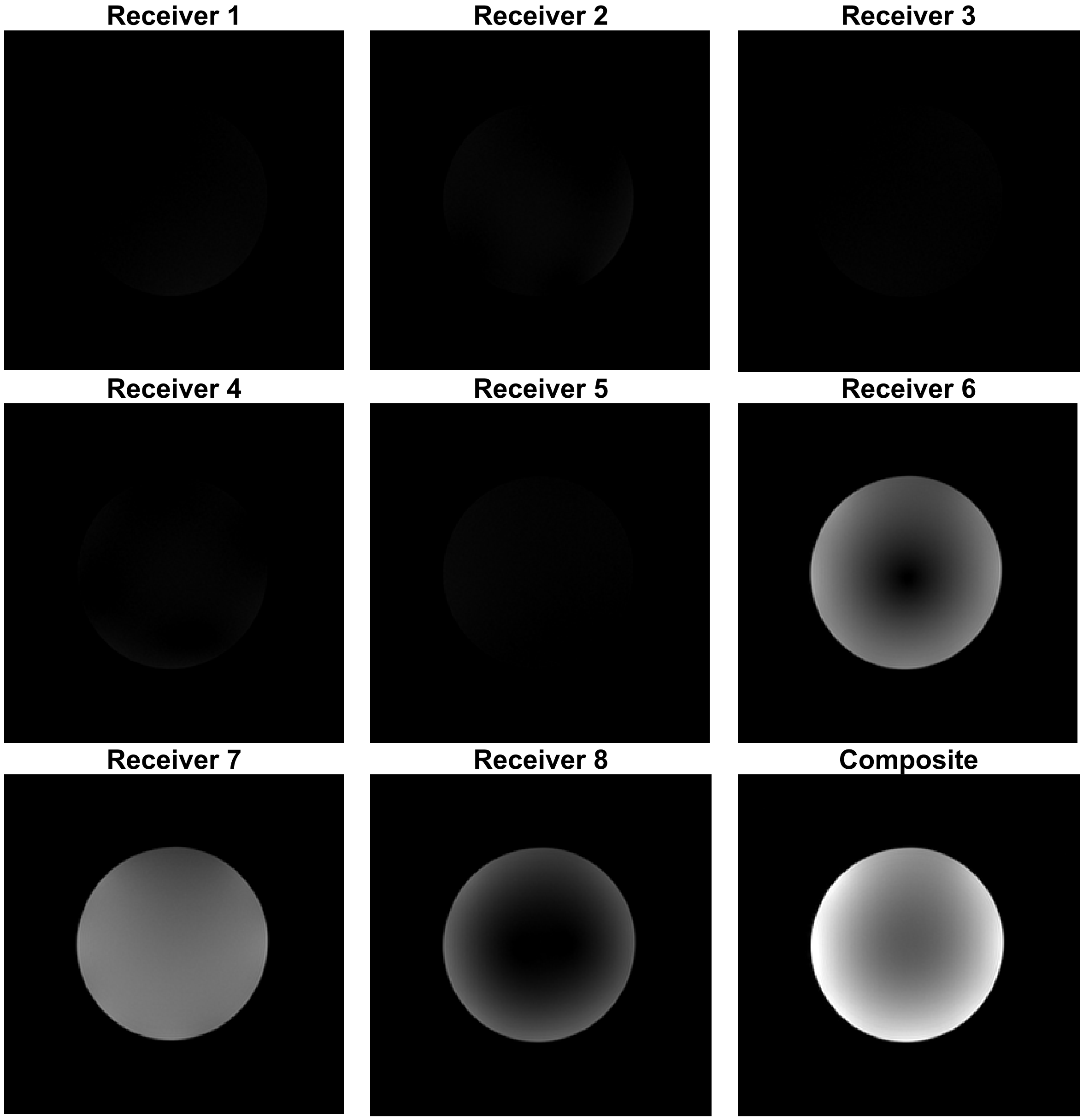


Figure 3-2-7-6 8NVARRAY Mode Signal Images Slice 3 (head)



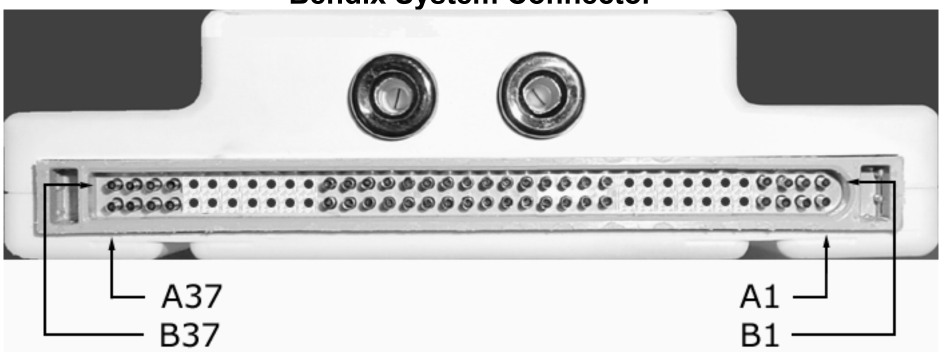
3-3 External Cable Check

Each MCP signal connection has a DC blocking capacitor in the cable connector. Therefore, continuity of the RF signal conductors cannot be determined in the usual way, by measuring from pin to pin using a multimeter. The external cable check must be performed in two parts, section 3-3-1 for the RF signal conductors and 3-3-2 for the PIN diode control conductors.

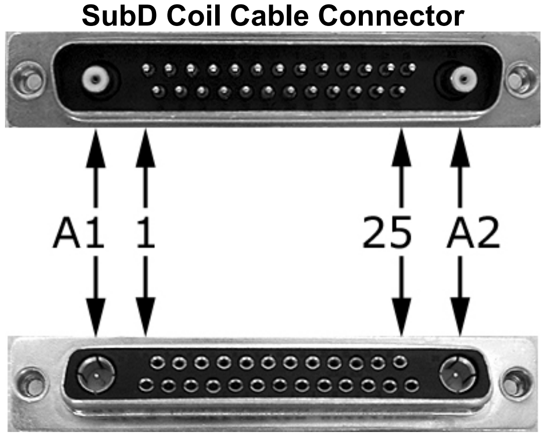
Use Figure 3-3 for pin identification while performing the procedures in sections 3-3 and 3-4.

System and Coil Connectors Pin Identification Figure 3-3

Bendix System Connector



SubD Coil Cable Connector



SubD Coil Connector

3-3-1 RF Signal Channels

The following procedure will rule out a short to GND for each of the 16 signal pins detailed in Table 3-3-1.

Note: This check does not determine continuity for each of the MCP signal connections.

Check the coil cable using the following procedure. Referring to Section 5-2, External Cable Replacement, disconnect the SubD coil cable connector. Using a digital multimeter, measure each of the 16 signal pins designated in Table 3-3-1 with respect to GND. Verify that the signal pins are not shorted to GND.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3-1

<table>
<tr><th>Signal Channel</th><th>SubD Coil Cable Connector</th><th>GND</th></tr>
<tr><td>MCP1</td><td>A1</td><td rowspan="8">2, 5, 7, 10, 12, 14, 17, 22, 24 and shield of coaxial contacts</td></tr>
<tr><td>MCP2</td><td>1</td></tr>
<tr><td>MCP3</td><td>16</td></tr>
<tr><td>MCP4</td><td>6</td></tr>
<tr><td>MCP5</td><td>9</td></tr>
<tr><td>MCP6</td><td>23</td></tr>
<tr><td>MCP7</td><td>13</td></tr>
<tr><td>MCP8</td><td>A2</td></tr>
<tr><th>Signal Channel</th><th>Bendix System Connector</th><th>GND</th></tr>
<tr><td>MCP1</td><td>B1</td><td rowspan="8">A12, A14, A16, A18,A20, A22, A24 and A26</td></tr>
<tr><td>MCP2</td><td>A2</td></tr>
<tr><td>MCP3</td><td>B3</td></tr>
<tr><td>MCP4</td><td>A4</td></tr>
<tr><td>MCP5</td><td>B34</td></tr>
<tr><td>MCP6</td><td>A35</td></tr>
<tr><td>MCP7</td><td>B36</td></tr>
<tr><td>MCP8</td><td>A37</td></tr>
</table>

If one or more of the pins are shorted to GND, replace the coil cable.

3-3-2 PIN Diode Control Channels

With the coil cable disconnected, use a multimeter to determine continuity between the designated pins for each control channel in Table 3-3-2. Also verify that the control pins are not shorted to GND.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3-2

Control Channel	SubD Coil Cable Connector	Bendix System Connector
MC 1	3	B12
MC 2	4	B14
MC 3	19	B16
MC 4	20	B18
MC 5	8	B20
MC 6	21	B22
MC 7	11	B24
MC 8	25	B26
+15 V	15	A25, B25
GND	2, 5, 7, 10, 12, 14, 17, 22, 24 and shield of coaxial contacts	A12, A14, A16, A18, A20, A22, A24 and A26

If one or more of the readings are > 3 ohms, replace the coil cable. If all readings are < 3 ohms, proceed to the **Diode Check.**

3-4 PIN Diodes Check

With the coil cable connected to the coil, use a multimeter to measure the diode drop across the coil connector pins detailed in Table 3-4. Use Figure 3-3 for pin identification while performing this procedure.

PIN DIODE EXPECTED READINGS – TABLE 3-4

Bendix System Connector		Reading
Positive Lead	**Negative Lead**	
B12	GND [A12]	1.6 ± 0.2
GND [A12]	B12	Open
B14	GND [A14]	1.6 ± 0.2
GND [A14]	B14	Open
B16	GND [A16]	1.6 ± 0.2
GND [A16]	B16	Open
B18	GND [A18]	1.6 ± 0.2
GND [A18]	B18	Open
B20	GND [A20]	1.6 ± 0.2
GND [A20]	B20	Open
B22	GND [A22]	1.5 ± 0.2
GND [A22]	B22	Open
B24	GND [A24]	1.5 ± 0.2
GND [A24]	B24	Open
B26	GND [A26]	1.5 ± 0.2
GND [A26]	B26	Open

If one or more of the readings indicate a short, replace the coil. If all of the above readings are correct, redo *3-2 Coil Imaging Performance Verification*. If the problem still exist, replace the 1.5T 8 Channel Neurovascular Array.

3-5 Mechanical Hardware Check

Not Applicable

SECTION 4 – MAINTENANCE

4-1 Coil Care

WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution.

The GE SIGNA 1.5T 8 Channel Neurovascular Array, Baseplate and Head Pads may be cleaned by wiping with a cloth dampened with a solution of 30% isopropyl alcohol and 70% tap water or a 10% Bleach solution.

4-2 Special Care Requirements

None

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 Disassembly of Coil

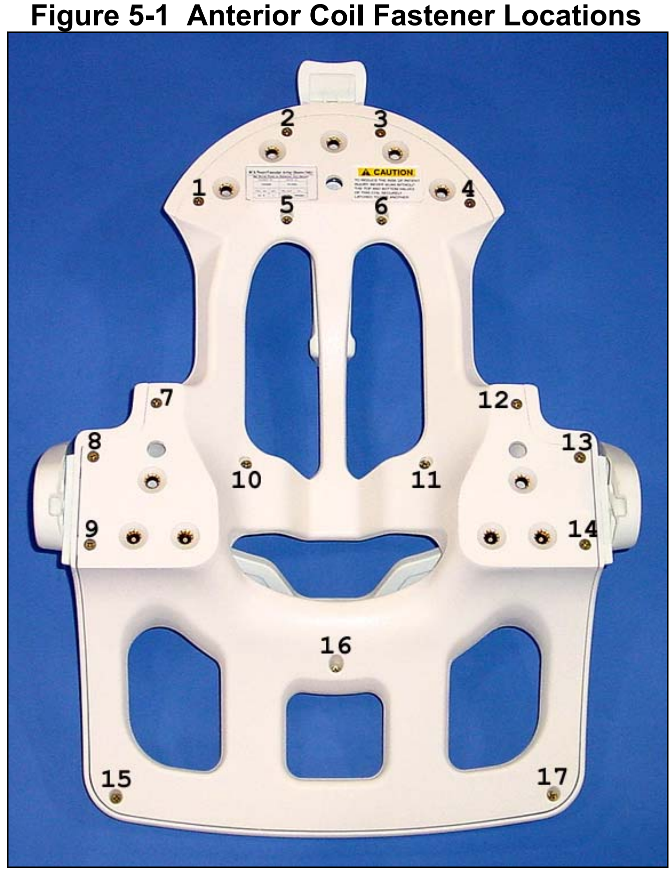
In order to perform the following procedures:

5-3-1 Superior Latch Assembly FRU Replacement
5-3-2 Right / Left Latch Assembly FRU Replacement

it will be necessary to remove the anterior coil cover from the anterior coil section.

1. Loosen, but do not remove, the 17 screws that fasten the anterior coil cover to the anterior coil section. The 17 fastener locations are detailed in Figure 5-1.
2. Carefully lift the anterior coil section away from the anterior coil cover.
3. Place the anterior coil section aside, being careful to keep the screws stored in their respective holes.
4. During FRU replacement and cover reinstallation, assure that the electronics in the anterior coil section are not disturbed.

Figure 5-1 Anterior Coil Fastener Locations



5-2 External Cable Replacement

Remove the 4 coil connector mounting screws located at the superior end of the coil. Remove the coil connector by pulling away from the coil in the superior direction.

Cables bearing this Coil ID label are compatible with software versions 10.X and newer. With software versions 11.X and newer, the system will generate an error when an incompatible coil is prescribed.

5-3 Mechanical Hardware Replacement

5-3-1 Superior Latch Assembly FRU Replacement

In order to perform the Superior Latch Assembly FRU Replacement, it will be necessary to remove the anterior coil cover from the anterior coil section. Follow the procedure detailed in section 5-1 Disassembly of Coil, then continue below.

1. Remove the 4 screws that fasten the superior latch assembly to the anterior coil cover. The 4 fastener locations are detailed in Figure 5-3-1-1.
2. Replace the superior latch using the Superior Latch Assembly FRU.

Figure 5-3-1-1 Superior Latch Assembly Fastener Locations

Figure 5-3-1-2 Superior Latch Assembly

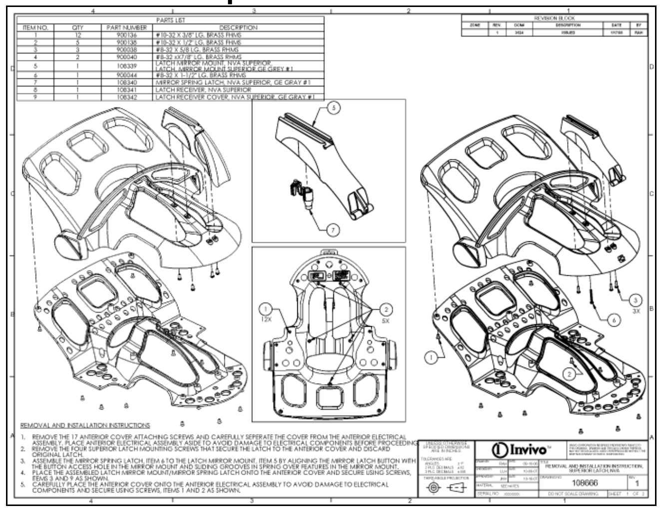
5-3-2 Right / Left Latch Assembly FRU Replacement

In order to perform the Right / Left Latch Assembly FRU Replacement, it will be necessary to remove the anterior coil cover from the anterior coil section. Follow the procedure detailed in section 5-1 Disassembly of Coil, then continue below.

Figure 5-3-1 Superior Latch Assembly FRU Replacement Instructions

1. Open the Superior Latch Assembly FRU package, and find the FRU Replacement Instructions, illustrated in Figure 5-3-1.

2. Follow the FRU Replacement Instructions while performing the Superior Latch Assembly replacement.



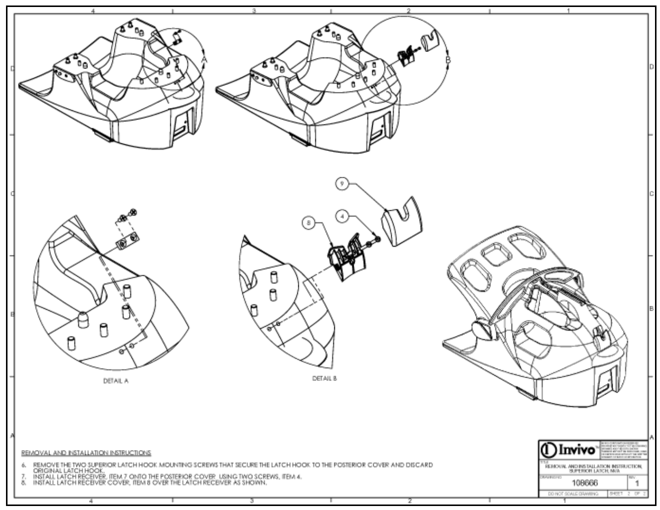
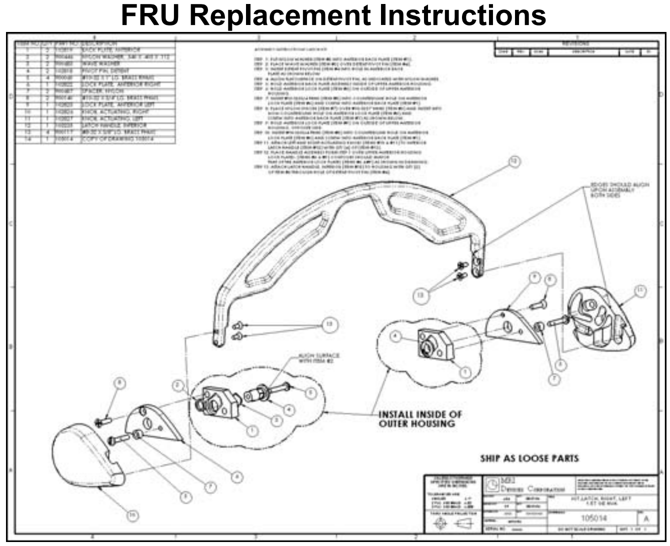


Figure 5-3-2 Superior Latch Assembly FRU Replacement Instructions

1. Open the Right / Left Latch Assembly FRU package, and find the FRU Replacement Instructions, illustrated in Figure 5-3-2.

2. Follow the FRU Replacement Instructions while performing the Right /Left Latch Assembly replacement.



SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS LIST – TABLE 6-1

Description	GE Part #	INVIVO #
1.5T 8 Channel Neurovascular Array	2317115-2	103045
Phantom Kit	2317115-3	103366
Cable	2317115-7	103025
Right / Left Latch Kit	2317115-10	105014
Superior Latch Kit	2317115-11	104975
17cm TLT Head Spherical Phantom	46-265826G6	N/a
10cm Spherical Phantom	46-317586G1	N/a
Large Cylindrical unified phantom	5342679	N/a
Phantom positioner, NV	5344806	N/a

Cables bearing this Coil ID label are compatible with software versions 10.X and newer. With software versions 11.X and newer, the system will generate an error when an incompatible coil is prescribed.

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Description	GE Part #	INVIVO #
Wedge Pad	2317115-8	101407
HNC Pad	2317115-6	102955

SECTION 7 – APPENDIX

7-1 Schematic

Refer to Block Diagram under Section 1-5 Theory of Operation.

7-2 Coil Configuration

Check the system coil configuration file for the 8NVHEAD and 8NVARRAY coil configurations. If 8NVHEAD and 8NVARRAY have not been installed, select 8NVHEAD and 8NVARRAY from the list of available coils, and install.

Revision History

Rev	Date	Author	Primary Reason for Change
A	10-24-02	L. Hyler	Initial Release
B	03-22-03	L. Hyler	Update QA Procedure – New Phantom and Positioner
C	03-24-03	L. Hyler	Refine Tables 3-3-1 and 3-3-2 Update Figures 3-2-1-3 and 3-2-1-4
D	03-24-03	L. Hyler	Change directory names in Sections 3-2-1 and 3-2-3
E	03-26-03	L. Hyler	Add mirror instructions to 2-1-1 Special Install Notes
F	03-26-03	L. Hyler	Update Service Methods CD in Section 1-3
G	03-31-03	L. Hyler	In 3-2-2, Change **snr_8nvarray_QA** to **snr_for_8nvarray_QA** Update Section 6, Renewable Parts
01	04-01-03	L. Hyler	
02	04-14-03	L. Hyler	Edit Table 3-2-3-1 Figure Reference Replace Figures 3-2-3-2 and 3-2-3-3 Edit 3-2-3 Plot Results, Item 5, to match Figure 3-2-3-2 Button
03	05-06-03	L. Hyler	Change 8NVHEAD to 8NVARRAY in Figures 3-2-4-4, 3-2-4-5 and 3-2-4-6
04	07-09-03 09-11-03	L. Hyler	Add Figure 3-2-1-5 and Correct Bottle Placement NOTE Update TABLE 3-2-3-2 SNR LIMITS 3-2-2.1 correct command to **snr_for_8nvarray_QA** Update 2-1 Coil Installation [8chNV install_rev1.doc]
05	10-22-03	L. Hyler	Corrected Operator's Manual Part Number in Section 1-3 Updated Loop Designations in Figures 1-5-2 through 1-5-5 Revised TABLE 3-2-3-2 SNR LIMITS Redesigned section 3-2-1 for both version 10.X and 11.X s/w
06	11-19-03	L. Hyler	Add Coil ID Information in Sections 5-2 and 6-1 Update Photos of Phantom Positioner
07	09-02-04	L. Hyler	
08	07-06-05	D. Visscher	Rebrand MRI Devices with Invivo (DCN 1378)
9	6/19/08	V. Thacker	Customer Change Request CCN0021 (DCN3994)
10	3/28/08	R. Hamilton	Change text and images to illustrate new superior latch and mirror. (DCN 3754)
11	6/24/09	R. M. Scott	Add Coil Configuration (DCN 5847)
12	10/29/09	M. Vaughn	Update per Change Request CCN0027 (Unified phantom set)